Pluriversal Approach to Co-Designing Delivery Robots with People with Disabilities

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Figure 1: Workshops with participants & examples of varied co-design materials generated using a pluriversal approach.

Abstract

On-demand, last-mile delivery – the transportation of goods from a local distribution center to the customer's door within a specified time window – is used by people with disabilities (PwDs) for a variety of reasons. While delivery robots have potential in this space, they often widen disparities in access for PwDs. Inspired by the concept of pluriversality, which embraces the diversity of worldviews, we facilitated participatory design workshops with PwDs to co-design accessible and equitable delivery robots. Our findings support the importance of delivery robots with varied form factors that are robust and customizable to support diverse PwDs and their respective needs and preferences. We also provide broader considerations for automation in delivery ecosystems.

CCS Concepts

• **Human-centered computing → Accessibility technologies**; • **Computer systems organization** → *Robotics.*

Keywords

delivery robots, accessibility, pluriversal design

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1 Introduction

On-demand, last-mile delivery (OD/LMD) - transportation of goods from a business to a customer's door during a time window specified by the customer – is used by people with disabilities (PwDs)[1] for various reasons. For some PwDs, delivery offers convenience. However, for others whose mobility is curtailed by limited accessible transportation options [45], OD/LMD represents a lifeline. Autonomous delivery robots (ADRs), or simply delivery robots, is an umbrella term [35] for systems that have the potential to bridge this gap. Companies tout several advantages of delivery robots for expanding the supply chain, including increases in efficiency; reductions in delivery times; the advancement of sustainability goals by championing 'clean', autonomous alternatives; and even the promotion of 'equity goals' by 'providing automation to more neighborhoods' [55]. Unfortunately, these autonomous systems have also created tension and barriers for PWDs (e.g. [12]). In the U.S. alone, examples of negative interactions with ADRs include a mobility scooter user who was repeatedly blocked from traveling on the sidewalk [20], a wheelchair user who could not use a curb cut that was blocked by a stuck robot [1], and a guide dog user who encountered problems with a robot [27]. These events have motivated new designs to ensure accessible interactions with PwDs [12, 30, 31]. These efforts primarily focused on people with mobility disabilities (PwMDs), necessitating input from other groups of PwDs. For example, delivery robots are often unable to detect the white canes of people who are blind or have low vision (pBLVs), decreasing safety. Still, these works centered on bystander interactions; one

[1] We use person-first ('PwDs') and identity-first ('disabled people') language interchangeably to reflect the diversity of preferences of disabled peoples and communities.

major ADR company reports that around 20% of their UK customers are PwDs [74]. Thus, it is imperative that ADRs are accessible for PwDs who are bystanders and customers. However, PwDs are diverse populations with varied access preferences and needs [71], making universal solutions difficult.

Pluriversality [23, 24, 53, 75] is a concept concerned with embracing plural universes that result from the diversity of humanity; it stands in contrast to universality, which typically seeks a single design that works for as many users as possible [50]. Pluriversal designers assert that there cannot be one universal solution and seek technologies that 'embrace ontological and epistemological diversity by being co-designed, co-produced and co-owned by the inhabitants of the socio-cultural territory in which they are embedded' [26, 29, 78]. We used a pluriversal approach [77] to participatory design workshops [49] with a variety of participants and sought potential solutions to answer the following research questions:

- **RQ1**: How can delivery robots be (re)designed to ensure that they are *accessible* and support a diverse range of PwDs with different needs?

To this end, we must interrogate who we are designing for and what their needs are. There is a dearth of research on motivations [57] and challenges facing PwDs regarding OD/LMD (e.g. [21, 52]). Furthermore, in the U.S., the poverty rate for PwDs is 2-3x more than that of people without disabilities [17]. Within the category of PwDs, subgroups are more likely to live in poverty, including people who are Black or Hispanic. Thus, we also consider:

- **RQ2**: What are the experiences, including motivation and challenges, that local PwDs face when receiving OD/LMD?
- **RQ3**: What are the desires of PwDs, who face intersecting social marginalization, for *equitable* delivery robots?

We recruited PwDs from different social backgrounds in our local community. Over two months, participants were invited to engage in two phases: an initial workshop aimed at capturing their experiences and perspectives on existing last-mile delivery services and robots toward (re)designing future delivery robots; and participatory prototyping workshops in which participants iterated on co-designs through in-person sessions and remote diary studies.

The contributions of this work are twofold: 1) Our work surfaced varied design ideas for delivery robots – different form factors, capabilities, features, and interaction modalities – to ensure that OD/LMD is accessible for a variety of PwDs' households; and 2) community-informed considerations that roboticists and designers should incorporate when designing ADRs.

2 Related Work

2.1 Autonomous delivery for PwDs

Many PwDs rely on public transportation, leading to challenges due to lack of adequate accessible options (e.g., [45, 70]). Thus, many PwDs use OD/LMD to get resources [21, 52], increasing exposure to ADRs. Current models include personal delivery devices using wheeled and legged robots that operate on the sidewalk and bike lanes [55], often referred to as sidewalk delivery robots (SDRs; e.g., Starship, KiwiBot, and Serve). Some use a mothership autonomous vehicle (AV) to transport SDRs to their delivery locations [35]. Similarly, some manufacturers are exploring autonomous aerial robots or drones (e.g. Amazon Prime Air [28]).

Most design research on ADRs has involved non-disabled people [6, 36, 39, 65, 79, 82], particularly in sidewalk interactions. However, sidewalk interactions with PwDs can be more challenging to navigate and have worse outcomes than those with a person who can easily detect and walk around a robot. Less work has involved PwDs to understand their perceptions and engage them in participatory efforts to ensure these interactions are accessible [30, 31, 83]. Recent work has begun focusing on sidewalk interactions between robots that operate in public settings and PwDs [12, 30, 31].

2.2 Pluriversal Design in HRI

Disability-related technology [81] seldom interrogates the social and material conditions of PwDs beyond their individual disability identifiers [71]. For example, PwDs in the U.S. are more likely to live in poverty [17] and experience food insecurity [18]. Within this category, subgroups of PwDs are more likely to live in poverty, including people who are Black or Hispanic and people with lower educational attainment [17]. However, in HCI and HRI, participatory design workshops often hinge on 'blue sky' ideation – imagination unbridled by social or economic constraints – which further marginalizes already under-served communities [32]. Previous work at the intersection of disability and racial, ethnic, and gender identity have illustrated how access needs for PwDs with intersecting social marginalization can be complex [13, 33]; however, these discourses seldom shift how participatory workshops are conducted in HRI (e.g. [80, 84]). Pluriversality arose from methodologies of decolonial design [2–5, 37, 41, 44, 75], which has been been used in other areas of HCI research to ensure technologies that are culturally and socially appropriate [8, 10]; these works demonstrate pluriversal approaches can enable the design of technology for the equitable interactions with under-served communities.

3 Methodology

Pluriversal design methodologies focus on fostering future possibilities for the diversity of humanity [53] and are guided by several design principles [77]. These are **(PD1)** using participatory approaches; **(PD2)** encouraging physical encounters (and creating 'safe' meeting environments to encourage discourse, which involves disagreement); **(PD3)** fostering alternative possibilities/solutions; **(PD4)** supporting connections between past and future through visual thinking and bodily expressions; and **(PD5)** practicing 'radical empathy'; **(PD6)** use of pluriversal narratives (to avoid distilling the findings). We aimed to implement these principles in our work.

3.1 Participants & Group Compositions

We recruited 14 local participants (9 women, 5 men) who were between 31 and 75 years old ($\bar{x}$ = 55, SD = 14.5; Table 1) via outreach to community centers, community email lists, local disability groups, and word of mouth. Although participants had varied disability experiences, demographic forms and participant statements during sessions indicated a predominance of visual, hearing, and mobility disabilities. Some participants specified using corrective technologies (i.e., glasses) while others did not (i.e. 'unspecified'). They reported varied employment and income statuses.

3.2 Methods, Study Locations & Procedure

We used a two-phase participatory design approach [49] **(PD1)**, which included an in-person **initial workshop** and **iterative participatory prototyping workshops**. All workshops were facilitated by the same two researchers. This study procedure was approved by our Institutional Review Board (IRB) and participants were compensated for their time. This work was conducted in Pittsburgh, a medium-sized U.S. city, over a two-month period in 2025. Participants selected a minimum of three accessible locations. Most of the selections were private meeting rooms in public libraries [22], offered because they are neutral democratic spaces **(PD2)**; only two workshops were held on our institution's campus. We used participant availability information to organize groups of between two to four participants. In some cases, cancellations resulted in solo sessions (see Table 1). Participants completed consent and demographic forms prior to in-person sessions, either via a secure web form or paper. Access needs were also obtained in advance and participants were asked to share relevant access needs during group sessions to ensure patience and empathy among group members **(PD5)**. Participants selected their own pseudonyms to allow control over how they were named and referenced in the data [40]. Income status categories are based on the number of people in a household [54].

3.2.1 ***Initial Workshop (W1).*** This phase focused on understanding the experiences of our participants with OD/LMD services and technologies **(RQ2)**. We wanted to understand how delivery robots and OD/LMD technologies could be (re)imagined and (re)designed **(PD4)** to ensure they are accessible and equitable for diverse groups of PwDs **(RQ3)**. There were three sessions: 1) *Focus group*. During this session, participants focused on their motivations for using OD/LMD, challenges they face, whether their experiences are impacted by their social identities (e.g., disability, race, geographic location, etc.), and their perceptions of OD/LMD technologies. At the end of this session, participants co-constructed a list of priorities for the future of accessible OD/LMD. Solo sessions were operated similarly to semi-structured interviews. 2) *Technology demonstration*. During this session, participants were introduced to delivery robots through videos with audio descriptions and tactile mock-ups. The videos featured Starship's wheeled sidewalk delivery robot (SDR) [73] and Agility Robotics' humanoid robot [59]. These two robots were selected to provide varied examples that included common, commercially available delivery robots (SDRs) as well as futuristic options (humanoids). 3) *Ideation Activity*. In this final session, participants engaged in various activities to help inform ideas and ensure that they were grounded in their experiences with OD/LMD. The first activity, inspired by somaesthetic reflection and articulation [42], called participants to reflect and express a detailed experience with OD/LMD; next, we provided a variety of modalities to support diverse preferences for communicating design ideas **(PD4)**, including verbally communicated and written scenarios [19, 61] and storytelling [14, 69]. We also offered 'generative toolkits' [62] that included materials for tactile ideation, such as Legos and other craft materials [51, 72], and audio recorders [49]. We also allowed participants to document ideas on their smartphones, which are common access technologies for many pBLVs (e.g. [63]). This workshop took 3.5 hours.

3.2.2 ***Participatory Prototyping Workshop (W2).*** We offered the option to also participate in up to three in-person iterative prototyping workshops. These iterated ideas that emerged in W1 **(RQ1)**. We incorporated a *design-led approach* to co-designing [62] accessible and equitable delivery robots with our participants/co-designers. W2 sessions involved a range of collaborative activities, such as engagements with a co-designed SDR mock-up that enabled participants to iterate on their ideas from W1. They also included high- and low-tech materials, such as technology probes (e.g., text-to-sign language software, sound buttons, audio cues) and the generative toolkits used in W1. After each workshop, participants engaged in a diary study for 7 days. This allowed participants to reflect on co-designs and suggest improvements for future workshops. While this phase focused on improving delivery robots, we allowed participants to envision alternate solutions to OD/LMD challenges **(PD3)**.

3.3 Qualitative Analysis

Qualitative data was analyzed deductively and inductively [25] throughout the study. Deductive codes were identified from prior work on delivery robots [30, 31, 65]. Inductive coding was performed by co-facilitators using reflexive thematic analysis [15, 16] in several steps. Reflexivity is the awareness of the bilateral influence of a researcher on the research setting and people [56]. After sessions, we wrote reflective memos (i.e., impressions of the session) that helped in writing analytic memos (i.e., evolving thoughts and interpretations) [11, 48]. The facilitators familiarized themselves with transcripts and generated initial codes; they met regularly to refine codes into themes that spanned the entire dataset. The entire research team met weekly to discuss emergent themes. Research findings were written as pluriversal narratives **(PD6)**.

4 Findings

We organized our findings around the experiences of PwDs with OD/LMD **(RQ2)** (see 4.1), followed by their desire for more accessible **(RQ1)** and equitable **(RQ3)** delivery robots (see 4.2-4.7).

4.1 Delivery Experiences of PwDs

4.1.1 ***Motivations.*** We identified several novel delivery motivations for PwDs. While these motivations pervaded all workshops, they were most prominently stated in W1. Our disabled participants had several overlapping motivations for using OD/LMD.

1) Inaccessible Infrastructure and Disability-Related Access. Several participants cited inaccessible infrastructure, their disability, and the lack of accommodations in physical stores as motivation for using OD/LMD. For example, iamcool and Mocha, guide dog users who live alone, reported inaccessible transportation; Mocha, who also lives in a high-rise apartment, also reported difficulties with her physical mobility. Astrid lives alone and described her symptoms of vasovagal syncope as including fainting, which impacts her mobility and requires seating. Thus, in-person shopping and carrying bags on the bus were infeasible and resulted

Table 1: Participant Demographics Overview

Pseudonym	Age Range[*]	Gender	Mobility Disability	Visual Disability	Additional Disabilities	Race/Ethnicity	Employment Status	Income Status	W1 Groups	W2 Groups
Pittsburghgirl	50–60	F	Arthritis	–	'Profoundly deaf'	White/Hispanic	Full-time	Extremely-to-Very Low	Group 1	Group A
Kassandra	30–40	F	AMC[1]; UED[2]; Arthritis	–	–	White	Part-time	Extremely-to-Very Low	Group 4	–
LGP	30–40	F	SB[3]	–	–	Asian	Volunteer	Extremely Low	Group 4	–
Mushy	55–65	F	Arthritis; UED[2]	Unspecified	Hearing Disability	Black/African American	N/A	Extremely-to-Very Low	Group 3	–
In-Remission-Man	65–75	M	Arthritis	–	Hearing Disability	Multiracial	Retired	Extremely Low	Group 3	–
Gabriel	65–75	F	Arthritis	Blind	–	White	Retired	High	Group 1	Group A
iamcool	50–60	M	–	Blind	–	Asian	Full-time	Low/Moderate	Group 8[**]	Group C
Effdi Jay Tee	40–50	F	–	Blind	Deaf	White/Hispanic	Unemployed	Extremely Low	Group 2	–
Abraham	30–40	M	–	Low Vision	–	White	Unemployed	N/A	Group 2	Group A
R.J.	60–70	M	–	Blind	Hearing Disability	White	Retired	Extremely-to-Very Low	Group 2	Group A
sleepsong	40–50	F	Arthritis; MS[4]	–	–	White	SSDI[†]	Extremely Low	Group 6[**]	Group C
Astrid	60–70	F	Arthritis	–	Hearing Disability	White	Part-time	Extremely Low	Group 7[**]	Group B
Mickey	70–80	M	–	Blind	–	White	Retired	High	Group 5	Group B
Mocha	60–70	F	Arthritis; UED[2]	Blind	Hearing Disability	White	Retired	Extremely-to-Very Low	Group 5	Group B

[1] AMC: Arthrogryposis Multiplex Congenita; [2] UED: Upper Extremity Disability; [3] SB: Spina Bifida; [4] MS: Multiple Sclerosis; [†] SSDI: Social Security Disability Insurance, which limits employment; [*] For anonymization, we provide decade ranges instead of exact age; [**] Solo workshop sessions.

in greater reliance on OD/LMD. Kassandra lives with her grandmother but requires more physical assistance than her grandmother can provide; furthermore, limited public transportation or caregiver assistance options on weekends led her to rely on deliveries for some items. Gabriel lives with her husband, but used OD/LMD because she wanted to shop independently despite limited options for human assistance during in-person shopping.

2) Survival. While related to the previous motivation, some participants commented on using OD/LMD as a lifeline. Effdi is Deaf-Blind and uses a white cane. She emphatically shared that delivery for her is about 'needing life-sustaining services' and that it is 'always [for] survival', citing inaccessible transportation options as a significant barrier.

3) Interdependent access. A few participants reported that OD/LMD provided opportunities for communal access to resources. For example, Mushy commented on using delivery to purchase 'snacks for [her] grandkids.' Pittsburghgirl uses OD/LMD due to her dual roles as a caregiver. She is the primary caretaker of her two children, with whom she lives in a 'food desert' (a neighborhood with limited grocery options [21]). For her, OD/LMD creates immediate access to resources, such as pharmacies and grocery stores. She also uses OD/LMD for supplies at her job as a geriatric nurse.

4) Independence. Several participants had empowering or positive motivations for using OD/LMD. Both participants who live alone (including LGP, a power wheelchair user, and iamcool) and those who live with family members (Gabriel and Kassandra) noted independence as a motivation for using OD/LMD.

5) Luxury and Comfort. sleepsong is a 'local disability advocate' who uses a power wheelchair. She typically orders food, for which her motivations include 'comfort' and familiarity – citing using delivery during times of physical exhaustion, chronic pain, and emotional duress. Moreover, Mushy and In-Remission-Man echoed ideas of OD/LMD as a luxury (e.g., to get themselves or their community something nice), and that they only use it when they have enough money.

6) Convenience, Ease, and reliability. Abraham and R.J. both live alone. R.J. uses a white cane and wears hearing-aids; and Abraham is low-vision. Both cited using delivery for convenience: Abraham having grocery stores in walking distance, and R.J. feeling comfortable using public transit to commute to his local stores. Moreover, LGP, Kassandra, Mickey, Astrid, and sleepsong also mentioned that they used OD/LMD for its reliability, convenience, and ease. Finally, Mickey, who uses a white cane and lives with his spouse in a private home, also cited the 'efficiency' of OD/LMD, especially since his spouse gave up driving due to aging.

4.1.2 ***Challenges.*** For PwDs, the failures of delivery persons to follow delivery instructions has adverse effects on their ability to get resources in a timely manner. Although participants cited issues with the *ordering* step of OD / LMD (e.g., digital accessibility), we focused here on the 'last steps' [58], namely issues with coordination of drop-off, performance of drop-off, and post-delivery.

1) Coordination of drop-off. For several PwDs, timing delays without notification are detrimental. For Kassandra, timing causes **inconvenience when physical assistance is needed** to retrieve items: 'I have to do things on a certain schedule because I depend on someone else to take care of me.' In addition, Mushy, who has a mobility disability, mentioned having to rush down when there are erroneous delivery notifications due to fears that her packages could be stolen; **theft concerns** were shared by In-Remission-Man, who often orders things to his brother's home instead of his own for security. Furthermore, participants reported that delivery drivers were unable to communicate through **accessible modalities**. Pittsburghgirl, who is profoundly deaf, noted that drivers tended to call despite her communicated access needs to text. Finally, some participants noted that experiences vary depending on the person delivering items. For example, LGP commented: 'the different drivers that come [...] change the way things are delivered and the communication or lack thereof with me.'

2) Drop-off. Drop-off issues depend on whether the delivery is attended (the recipient meets the delivery person) or unattended (the package is left at a designated spot) [58]. For pBLV customers, failures by **attended** delivery people to meet in an exact location can pose unique challenges; for example, Effdi and R.J. cited physical safety concerns with approaching an incorrect vehicle. Abraham, who is low vision, cited challenges with identifying drivers who assume he can see them. Moreover, these interpersonal interactions can result in discriminatory encounters: R.J. described **ableism** by delivery people who tended to ignore him – a sentiment with which Effdi and Abraham agreed. Participants also cited failures by delivery people, who can notify them either through designated communication channels (e.g., apps, texts, calls) or via intercoms, to do so upon package arrival. iamcool also cited language barriers: 'drivers don't always speak English well, so when I need to ask questions [..] Sometimes [...] I don't get the answers.'

As identified in prior work [43, 52], our BLV participants reported difficulties finding and identifying **unattended** packages that were not placed in the *specified* or *designated* delivery locations. Incorrect locations can be **physically inaccessible**. Kassandra shared that delivery people do not read instructions that specify a closer, more accessible location; instead, they tend to leave items in her inaccessible mailbox. Several participants noted that incorrectly placed items can cause **bodily harm** by being tripping hazards. Furthermore, Effdi and iamcool added that difficulties in locating misplaced items are exacerbated by inaccessibly labeled packages [43].

3) Post-delivery. Participants discussed a range of post-delivery issues, such as incorrect and poor quality items, problems with proof of delivery, and inaccessible customer support to correct problems. For many PwDs, issues with the earlier 'last steps' of delivery **forces them to depend on non-disabled people** [43]; however, most of our participants live alone, reducing opportunities for assistance. LGP noted that while some delivery people are patient to help her verify her items prior to departing, this varies significantly. For several participants, opportunities for post-delivery support were fraught and tended to be biased toward inaccessible mediums. iamcool, who is blind, cited challenges with customer support asking, 'Can you find somebody to help you? [...] send a picture of this' noting 'its not as easy as they think'; this reflects a **bias towards visual proof** to resolve delivery issues. Kassandra, who has limited upper extremity mobility, noted that taking pictures is difficult and requires a specialized setup, which she does not have at home. In general, the performance of delivery varied by the delivery person and perceived differences in labor practices that prioritize expediency. Many participants were empathetic about these circumstances facing delivery people; however, their amount of consideration for these extenuating factors also depended on their motivations for delivery (see 4.1.1).

4.2 Accessible Communication

Communication is imperative for the 'last-few-steps' [58] of delivery. These include coordination, drop-off, and post-delivery, which necessitate timely and accessible communication through notifications delivered by text, call, or app.

1) Communication To Delivery Robots. Participants in the workshops showed preferences for various ways of communicating with delivery robots. **App-based interfaces** were a common option that several participants suggested. Communication through an app on their smartphone was preferential for some participants due to accessibility, particularly for people with hearing and visual disabilities. Also, several participants suggested non-smartphone options for people who are not proficient with them.

Tangible interfaces, such as tactile interfaces (e.g., tablets) and clearly marked buttons, were cited as common input to enable interaction with diverse range of customers. Participants shared the importance of on-body inputs to enable easy interaction with delivery robots by non-high-tech customers without smartphones, which Pittsburghgirl noted would be useful for sending resources to members of her community; this idea was also echoed by sleepsong, who touted the benefits of 'redundancy' (W2). Furthermore, tactile or touch-sensitive input could activate an alarm to deter bystanders from attempting to take items from the robot while en route (Gabriel, W2). Finally, some participants also showed interest in the use of biometrics to unlock the robot's grocery compartment.

Participants also suggested **verbal communication** [30, 31]. Many participants were guide dog users and wanted delivery robots to be similarly capable of following them to their apartments for drop-offs. Robots that support diverse human communication via voice interaction and natural language capabilities should account for diverse accents, including disability-specific speech variations, the latter of which Pittsburghgirl stressed. Also, Pittsburghgirl, who uses sign language, expressed 'I would love to see [...] an ASL robot' (W1), communicating the desire to communicate with delivery robots in **gesture-based languages**. Participants also expressed desires for **interoperability with Augmentative and Alternative Communication (AAC)** [9], which can be 'no-tech', such as observing facial expressions; or 'high-tech', such as speech-generating devices and text-based interfaces. Gabriel noted, 'There's just like a puffer tube. [...] Some of the commands could be like unlock, lift, slide if it's coming out to the side.'

2) Communication From Delivery Robots. Just as accessible communication is important for bystanders (e.g. [31]), it is also critical for customers. Several participants suggested various modalities for communication. They shared that **visual cues**, such as signal lights and flags, can be helpful for low vision customers to ensure rendezvousing with a delivery robot. Similarly, diverse output modalities, such as *gesture-based* and **text-based** communication, are also critical to support diverse local communities. Participants also suggested **auditory cues**, which include speech and non-speech. Gabriel and Mocha envisioned delivery robots indicating their location to BLV customers by producing auditory cues similar to those emitted by electric vehicles (EVs) or an Air Tag. R.J. added that specific verbal instructions and directions from the robot could further assist with locating it (e.g. [64]).

Some participants discussed their preferences for **speech characteristics** of the delivery robot. While Mickey had preferences for a human-like ('realistic') sounding voice, Astrid preferred a computerized voice. She shared that 'I think after [delivery robots] have been around for a while, people [will] hear the computerized voice [and think] "oh, that's the [delivery] robot"'. Astrid, who also has a hearing disability, and Mickey suggested that sounds must be **automatically-modulated** based on ambient noise to ensure they are perceivable by people with hearing disabilities (W2).

4.3 Customizable & Personalizable Interactions

Participants articulated their desire for personalized and customizable interactions with delivery robots that were considerate of disability-specific as well as social and cultural factors.

4.3.1 ***Disability-Awareness: Proxemics and Notions of Space***. Proximity in bystander interactions with PwDs [12, 31] has been explored with respect to importance of right-of-way for people using mobility devices/aids. These notions were echoed by several participants, who also considered bystander interactions for PwDs. Kassandra noted safety issues: 'If [the robot] didn't detect someone in [...] a manual wheelchair, they could potentially push that person into the middle of the street' (W1). Participants also shared the importance of **close proximity interactions** for people with sensory disabilities, including people with hearing disabilities and pBLVs – an idea that R.J. evoked while exploring the distances from which he could hear specific sounds. Close interactions were seen as critical for **facilitating handovers** with PwMDs for whom reaching can be difficult. Kassandra noted: 'a person in a wheelchair has to get closer to the robot to be able to scan their phone and grab things out'. Heavy items were also highlighted, with sleepsong noting: 'I can't carry bags very far [...] If I had a robot delivering stuff, I would not be able to take care of the delivery on my own. I'd need someone to help me [...] put stuff away'(W1). Thus, many people with limited mobility shared ideas about the robot helping with post-delivery processes, such as putting items away (see 4.6).

Several participants discussed **height considerations** for SDRs, in particular. LGP noted that there is variability in the heights of wheelchairs, which delivery robots do not take into consideration. Similarly, sleepsong suggested that customers be able to adjust (i.e. **customize**) robot height through voice commands or buttons to enable customers in wheelchairs to retrieve their items. iamcool added the idea of **personalization** of automated height adjustments for specific users depending on wheelchair height.

4.3.2 ***Socially and Culturally Appropriate Behaviors.*** Participants suggested that delivery robots should be capable of socially appropriate behaviors. This is most pronounced in *social navigation*; for example, Mickey suggested delivery robots adhere to foot traffic conventions, such as waiting for people to pass; he encouraged the use of 'politeness' as it 'encourages people a little more' [34]. Furthermore, he suggested that robots communicate their directionality, which is important for pBLV passersby: 'I'll be on your right or I'm gonna be on your left' (W2). Similarly, others suggested that delivery robots be aware of cultural norms. For example, iamcool evoked the idea of delivery robots 'taking off' or cleaning off shoes before entering households, which is common across many cultures. Although iamcool was uncertain about 'how effectively a robot can use' a doormat to wipe its wheels, he suggested the robot respond to verbal commands, such as to wait while he cleans its wheels prior to its entry. This latter idea is important for post-delivery assistance (see 4.6).

4.4 Delivery Safety and Malfunction Support

The participants shared ideas on the *safety of their deliveries*, which depended on the type of delivery. Some participants discussed safety measures for unattended deliveries, such as SDR compartments having a **detachable lockbox**, which could be left at a specified location near the user's residence. Moreover, *cost* and *sustainability* prompted them to consider how robots could **interface with existing lock-boxes** at the front of residences. For attended delivery, participants preferred that the robot compartment remain locked. They speculated on high- and low-tech options to unlock the robot compartment, including near-field communication (NFC) between smartphones and the robot or passcode via keypad or touchpad on the robot's body. Regardless of delivery paradigms, participants urged that compartments must be insulated to maintain the quality of medications and food items. They also noted that robots should either **automatically sanitize compartments** or have covers to prevent contamination by allergens or germs. This was surfaced repeatedly by Pittsburghgirl, who works as a geriatric nurse to immunocompromised adults, some of whom have allergies.

Participants discussed **alarm mechanisms** to deter bystanders from vandalizing 'or meddling' (Mickey) with ADRs en route. iamcool, who shared that he would allow the robot to enter his home while he was away, discussed safety precautions for preventing burglars from following the robot into his home: 'A robot can have a camera and things like that, you know. If the law allows it, [the robot] can electric shock that person [...] [or use] bear spray.' Participants also discussed robot malfunctions en route, near, or in their residences. iamcool suggested 'robot roadside assistance' provided by the robot's company. LGP echoed similar ideas. Kassandra added the importance of hasty deployment of technicians if the robot fails en route to ensure package delivery or refunds. But she remarked that **transparency with the customer** (e.g. [68]) about the origin of the problem is critical because it may suggest routine flaws.

4.5 Accessible Robot Features & Form Factors

Our participants offered significant insight into the capabilities of different mechanical systems due to their experiential knowledge with their own or community members' mobility devices. For example, sleepsong, LGP, and Kassandra - all PwMDs who use power wheelchairs – were acutely aware that SDRs had severe limitations due to the poor sidewalks and steeply sloped neighborhoods in our city. Additionally, sleepsong shared that many SDRs would get stuck in cobblestone sidewalks in her neighborhood due to their small wheels (W1). As such, several participants suggested *environmentally appropriate* features for delivery robots, including all-weather, high 'traction' wheels, and winterized bodies for operation in inclement weather. Furthermore, our participants were keenly aware that: 'depending on the house, depending on the type of disabilities you have, you can have several different shapes or models of [delivery] robots' (iamcool, W2). They offered several features that they considered *architecturally appropriate* to perform delivery tasks in various types of buildings. The participants brainstormed legs, different chassis configurations (Abraham and Gabriel in W2), and tri-wheels [66] (Astrid, W1) for climbing stairs or ambulating on uneven terrain (Fig.1B & C).

Features recommended by the participants, particularly types of motions, suggested other form factors beyond those of SDRs. For example, **delivery drones** could be beneficial when faced with a *lack of appropriate transportation infrastructure* that a SDR could use – a point that Mickey added due to the lack of sidewalks in his suburb; and by Abraham, who speculated about the potentials of delivery drones or legged robots in *remote areas*. Mocha, who lives in a high-rise apartment suggested that 'a drone [...] could hover at the window and I could take the food out', or land on a customer's balcony (W2). Moreover, she speculated that delivery drones could be more efficient, particularly for the fast delivery of 'food items.'

Despite the utility of this form factor, participants also considered the **limitations of delivery drones**. The first they cited was 1) *Payload*. Mocha shared that 'a 40 pound bag [of dog food] [...] might be too much for a drone. But [...] it could bring smaller packages' (W1); 2) *Safety considerations*. In a diary reflection, Astrid questioned whether grocery and food items delivered by drone would be safe from **theft**, assuming that unlike an SDR, they would have no locked compartment; 3) *Privacy*. Finally, Effdi cited concerns about privacy when using delivery drones: 'The big thing for me is privacy[...] It's the public seeing everything when you [are] just trying to live your life [...] just bring it to my door'.

Furthermore, participants expressed their desire for robots to **press** the intercom and elevator buttons and **knock** on their doors upon arrival, the latter being a social norm. Thus, they discussed delivery robots having arm and hand-shaped appendages, a design feature that can also facilitate accessible handovers. For example, Pittsburghgirl suggested that the robot have 'retractable' arms for providing physical assistance by placing items on the lap of PwMDs who are wheelchair users. In addition, sleepsong suggested a '**hydraulic lift** that can lift your order up to you', a design feature that she likened to one that her power wheelchair has; the latter idea was also represented in the co-designed mock-up in W2 sessions. Finally, participants shared preferences for the **sizes of delivery robots**. Participants were concerned about the **carrying capacity and weight limits** of delivery robots. For example, R.J. and Effdi (W1) were not enthusiastic about their space and weight limitations, which they deemed too restrictive to meet their needs. However, the **size of the robot should depend on the environment**:'I like the smaller size of the robot because it's less of a safety risk' (Pittsburghgirl). Furthermore, participants suggested *compact robots* in settings with limited sidewalks or narrow roadways.

4.6 Delivery Robots & Their Roles

Several participants were excited about the opportunity for delivery robots to provide them with post-delivery assistance. The humanoid robot form factor evoked fear and discomfort in a range of participants (mostly those who could visually perceive it), leading most participants to ideate around SDRs, which they viewed as more friendly. However, for participants, for whom the humanoid form factor mirrored that of a human caregiver, they noted its utility. For example, Kassandra, who orders when she has a caretaker to bring things in and pack them away, preferred a humanoid delivery robot; she speculated that it 'would probably be better since I'm not able to physically reach [...] and grab things.' Furthermore, iamcool brainstormed a delivery robot with a **barcode reader** to scan items as he removes them from the compartment to help him *identify and confirm* delivered items. He described the capability to *query the robot for additional information* about each item (e.g. cooking instructions) (W2). Moreover, Mickey brainstormed about the delivery robot being able to **return items** to the store – an idea that was echoed by several other participants. However, in Mickey's idea, he envisioned the delivery robot operating like a **shopping assistant** that could give him feedback on his visual appearance when trying on new clothes (W1); the robot could also **automatically refund** money for items that he did not plan to keep. Finally, iamcool suggested the benefits of a delivery robot helping him with assembly of new furniture, either with the construction or with helping to *identify items and where they fit in the assembly*: 'if the robot can help me [with assembly] or orally, even that would help. But if it can do it by itself, that's nice too. That's another feature' (W2).

Some participants also contemplated the *ethical ramifications of delivery robots that provide auxiliary support*. For example, Pittsburghgirl offered an idea for delivery robots en route informing people with visual disabilities when to cross the street (W1), presumably when audio crossing signals are not working/available. However, based on her lived experience, Gabriel cautioned against delivery robots telling pBLVs to cross the street if *the crosswalk signals were not working and the robot misinterpreted this* – citing fears about intelligent systems providing incorrect information (i.e., AI 'hallucinations'), that could be life-threatening.

4.7 Social & Ethical Tensions

Some participants were **optimistic** about delivery robots, which they perceived as being more morally trustworthy [47] than delivery people; some of this is rooted in their negative experiences with delivery people (see 4.1.2). For example, Effdi shared 'The robot is not prejudice[d]. The robot is not ableist, as the robot does not judge. The robot [gives] you your stuff and [goes] on its way. That's all I want' (W1). However, some participants were **skeptical** about the potential of delivery robots to disrupt delivery employment for vulnerable populations. For example, sleepsong had previous experience advocating about the detrimental impacts of premature deployments of SDRs on sidewalk access, particularly for local PwDs. For her, improving the accessibility of delivery robots was a means of *harm reduction*, particularly for members of the disability community: 'I just don't like them at all, but I recognize they're going to happen [...] so I might as well make sure they're as accessible as possible'(W1). sleepsong's skepticism enabled her to write a counter-narrative [14, 69] (Fig. 1D, W2) in which she articulated critical steps that delivery robots need; that is, **handling complex interactions with humans**. sleepsong ended her story with her desire to have delivery people instead of robots.

The tensions in values and experiences between participants yielded fruitful discourse. For example, sleepsong and iamcool, who have different perspectives and lived experiences with disability, were in a group together; this allowed them to confront these tensions. For example, iamcool shared, 'I just want to be safe. I want my things delivered [...] I don't trust people coming in [when] I can't see what they're doing [...] I have no choice.' He added that he has 'no issue whether it is a delivery person versus a robot' as long as he 'can always find [his deliveries at a] consistent location'(W2). For some PwDs, the use of delivery robots can provide opportunities for consistent and reliable access to life-sustaining resources, especially since there is so much variability in delivery people. Participants showed interest in **mixed human-robot delivery paradigms**. These discussions prompted further considerations: 'I think depending on *why* [companies] are using robots [for example, whether it is] just really to save money for delivery [by] not having the delivery person [...] maybe the *robot can carry things* from the truck to the residence [...] there could be some *people with bad knees [who] can still drive* [...] but the robot can actually carry things' (iamcool, W2). Here, these tensions fostered discourse around paradigms that can support disabled workers; however, more work needs to be done to understand the unique challenges and desires of disabled workers. Still, there are roles where human assistance will remain paramount. For example, Gabriel and Pittsburghgirl shared their **preferences for human customer service** as compared to chatbots. Both of them repeatedly stated their preferences for humans, who are capable of *empathy*, saying for example: '[a] person could tell you what food tastes good because they actually eat it. The robot doesn't know' (Pittsburghgirl, W1).

4.7.1 ***Delivery Robots & Low-Income Communities.*** Our participants hailed from different neighborhoods but a common factor is that many lived in **low-income communities**. For example, sleepsong commented on her discomfort with neighbors observing her receiving items from a delivery robot 'to have some [fancy] robot bringing me [a package] instead of having a person [...] no, I don't want that' (W1). For residents of low-income neighborhoods, high-tech, such as robots, can conjure negative associations (e.g., impending gentrification [38] and divestment from public resources towards private ones [12]). To counter this, Mushy and In-Remission-Man stressed the importance of **gradual introductions** of delivery robots, citing diverse and fearful social perceptions of unfamiliar high-tech systems. For example, Mushy quipped that while Digit, the humanoid robot would be antagonized, Starship's SDR would be perceived as less threatening. Similarly, In-Remission-Man observed it may just prompt 'curiosity'. He suggested that delivery robots could be introduced gradually to communities, offering potential venues like **senior centers and libraries** where people can 'watch [delivery robots] in action so that they can be familiar with it before they see it.' He added that it is particularly critical for vulnerable communities: 'You got to introduce [delivery robots] to the people that will be affected the most.'

5 Discussion

It is crucial to consider the varied experiences PwDs have with OD/LMD. Previous work on OD/LMD focused on use among non-disabled populations (e.g. [57]). We affirmed that many PwDs use OD/LMD for reasons related to accessibility [52]. Majority of our participants face additional socioeconomic hardship [17] and live in low-income communities with few resources [18]. OD/LMD connects them to valuable and life-sustaining resources [21]. Our participatory design workshops helped surface these motivations and challenges, which were critical in helping our disabled co-designers to (re)design delivery robots to better support their needs and the needs of their communities.

We summarize practical insights for future delivery robot design and incorporating pluriversal approaches to robot design for the broader HRI community, especially when working with PwDs. *(1) Recommendations for Future Delivery robots.* **(A) Accessible communication modalities.** We uncovered varied and inclusive communication modalities our disabled co-designers suggested beyond voice interaction [31]. Participants suggested tangible inputs, such as buttons, to ensure access amongst their low-tech community members; they also suggested *gesture-based inputs* [7, 67], and interoperability with *AACs* (e.g [9, 76]), both of which are underrepresented communication modalities in robotics. **(B) Diverse Form Factors & Features.** Unlike in previous work [30, 31, 46], we enabled participants to (re)design delivery robots to have a variety of form factors and features to ensure robots that support their *accessibility needs*. Participants considered the pros and cons of certain form factors with respect to their *local infrastructure and residences*; they described different tasks they wanted delivery robots to perform in a range of environments and envisioned delivery robots having features that would allow them to *climb stairs and press elevator buttons*. While many companies operate homogeneous ADR fleets, this work demonstrates that PwDs recognize the utility of fleets composed of robots with different form factors. Companies and researchers could rely on user feedback about disability-access needs from PwDs and crowdsourcing infrastructure and residential accessibility [60] to assist with selective deployment of delivery robots with specific form factors. Still, participants were conflicted about larger delivery robots for bulk shopping to avoid additional fees incurred by frequent deliveries versus bystander safety.

(2) Pluriversality in Robot Design. This is done by embracing: **(A) Tensions in the design process.** Pluriversal approaches enable co-designers, including researchers, to respect differences in lived experience [53], including disability and other social factors; as such, our disabled participants were able to have fruitful discourses about how these different perspectives can yield equitable interactions with delivery robots (see 4.7). They discussed social and ethical considerations for delivery robots in local communities to ensure equitable introductions that do not further ostracize PwDs, many of whom live in low-income communities (e.g. [32]). **(B) Centering affected PwDs' lived experiences.** PwDs can provide valuable suggestions for accessibility features that may support other PwDs with different disabilities. Our participants in mixed disability workshops were able to provide additional ideas about features that could support other group members (see 4.3.1). However, their suggestions could be limited. In previous work, PwMD and non-disabled co-designers envisioned delivery robots that could provide *auxiliary assistance* to pBLVs crossing the street [31]. Although a PwMD echoed a similar idea in our work (see 4.6), a pBLV noted a robot making erroneous suggestions could be life-threatening. **(C) Researcher Reflexivity.** We encouraged participants to envision solutions to their OD/LMD challenges beyond delivery robots, which is central to pluriversal design [77]. Here, *researcher reflexivity* is vital [77] as it encourages researchers to be aware of their own biases, which may be technology-oriented, and to defer to the needs of community members.

6 Conclusion

In this work, we explored how PwDs would (re)design delivery robots. We used a pluriversal approach to conduct a series of participatory design workshops with diverse local PwDs. This resulted in diverse design recommendations, including varied communication modalities, delivery robot form factors, and different environmentally- and task-specific features to meet the needs of PwDs. This work also demonstrates that design engagements that end with disparate results are not flawed – they represent the diversity of the human experience.

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